AI Integrated in Society: in China vs US

BLUF

We predict that the upcoming generation of military personnel from our most prominent near-peer adversary—the PLA—will be better prepared than future US military to utilize the full capabilities of GAI upon Force entry. China's emphasis on mandatory GAI curricula and early adoption of US and homegrown GAI technologies have and will continue to put Chinese ahead of US students in this space before they become our recruits. To understand how we can best support Airmen in closing this proficiency gap, we must first understand China's current progress and future vision for educating the next generation in GAI.

Highlights Preview

>> Zhejiang, China implementing both education and application of generative AI in primary and secondary schools. Compulsory, with 1,000 experimental institutions by 2025 (see Education).

>> Intellectual Property (IP) output in China 3x that of US (see Appendix B below)

>> Each major US company that has a technology in this space has a Chinese equivalent. These Chinese companies have either developed at least one generative AI technology or have invested in one (Appendix A).

>> Both the US and China investing in military applications of AI, though none suggest robust generative AI adoption in the military yet. Research, risk analysis under way in both countries (see Military).

Education

Since 2018, China's Ministry of Education mandated AI curricula in its high schools[1]. Many of China's provinces have swiftly transitioned in this mandate by implementing GAI education in primary and secondary schools since 2019, paving the way for more ambitious policy. These policies refer to (1) utilizing GAI tools to enhance classroom lessons and (2) teaching a curriculum on GAI itself (analogous to Information Technology or software development studies, for instance). Besides classroom implementation, China also boasts (3) education-centered, publicly available GAI tools. Here we provide a brief breakdown.

(1) Lesson enhancement: Some Chinese classrooms already feature some of the capabilities in our ideated technologies. Teaching assistant bots offer personal guidance based on each student's individual performance, generate curricula/course content, and hold conversational lessons with students.[2] Instructional use of GAI in the US, however, is institution-specific. Education experts offer nuanced, situational, and subject-specific guidance[3].

(2) Curricula: Local and province-wide educational policies have already taken root, affecting millions of Chinese students at primary and secondary schools across the country. In March 2023, Zhejiang[4] announced that AI core curricula would become compulsory in primary and secondary schools[5] with 1,000 experimental institutions by 2025[6] . It would build form the over 700 institutions already constructed up to that point. Cities such as Nanjing[7], Shanghai and Beijing[8] have launched their own programs as well. Renming University's affiliated high school located in Beijing, for instance, launched its "Artificial Intelligence + X" curriculum, which includes machine learning, deep learning, data mining, automatic driving, natural language processing, and computer vision. Information Technology and AI are even integrated in many of these cities and provinces' versions of the national standardized test (gaokao).

(3) Educational tools: Chinese students already have access to personal-use educational GAI tools. A number of schools in China have already implemented a technology resembling our proposed tutor bot Squirrel AI[9][10] to help struggling students[11] and to reduce student burden[12]. A GAI (Wenku) on China's most popular browser (Baidu) offers chat assistance and multimedia content generation[13] , often used for academic assignments.

<u>Military</u>

Policy: Both the US[14] and China have outlined policy to make AI trustworthy before full-scale implementation. While a lack of concrete policy appears to hinder AETC's ability to use GAI, China promotes AI standardization and application in the military[15] .

Technology: As of this report, the US military has assessed a small variety of LLMs and chatbots. Different branches have sanctioned use of some for certain CUI applications. AETC had for a time approved use of ChatGPT+ and AskSage AI (an extensible GAI platform hosting a number of industry-leading LLMs, including GPT 3.5 and 4). In June 2023, the Navy approved a conversational tech support chatbot called Amelia[16] . As of this report, we have not identified Chinese military-specific GAI technologies.[17] Their military, however, is also ideating potential use cases and applications, such as for staff assistants[18].

Careers: Several Chinese military academies list AI-centered careers on their job ads[19]. They range in recruitment, algorithm engineer, and software engineers ranging from recent graduates to experienced professionals. The US military's desire to explore and adopt GAI outpaces our preparedness; as of writing, there are no publicly-listed AI careers in any of the military branches advertised[20].

Summary

Industry leaders continue to be an insightful proxy to understand broad technology trends and capabilities by country. We can extend this public information to help develop domestically as well as infer the capabilities of peer nations. In the context of GAI, although conceptual work and group efforts can be traced back before the pandemic, we notice a surge in activity even among leading companies post-2020, both domestically and abroad. China's relatively earlier posturing as an AI-fluent society, however, has positioned its government and companies to output and integrate the technology in society sooner and faster than the US. However this reflects integration in the two nations' militaries and interaction with each other will depend on perceived threat or potential for partnership with the other.

APPENDIX A: Industry

Below is a summary of the latest AI technologies from leading companies both in the US and China. Similarly color-coded companies indicate the rough equivalent between the two countries' companies (for instance, Alibaba is China's Amazon). The **bolded** descriptions and **highlighted** technologies and their descriptions indicate a generative AI technology.

Company	Technology	Modality and Application
AskSage, Inc.	**chat.asksage.ai**	secure access to (**text-based**) GAI models designed for gov and commercial use
Aimoshu	**chat.aimoshu.cc**	Access to (**text, illustration, multimodal**) GAI models
Khan Academy	**Khanmigo**	**Text-based chatbot**
Khan Academy	(see above)	(very limited accessibility in China)
Squirrel AI	**Intelligent Adaptive Learning System (IALS)**	**Text-based chatbot (in over 1200 Chinese schools)**
X (formerly Twitter)	**LLM**	**Text-based model (under development)**
Weibo	**"painting" service**	**Text-to-image**
TikTok (ByteDance)	**AR effects, virtual assistant**	**Visual (image, video), text**
Douyin (ByteDance)	**Copyright-protected AIGC system software, AR effects, virtual assistant**	**Visual (image, video), text**
OpenAI	**ChatGPT**	**Text-based chatbot and developer API**
	DALL-E	**Text-to-image and developer API**
	Whisper	Voice-to-text and translation, developer API

Zhipu	15 AI-based products listed, at least **10 in operation**, all appear to have some generative component	**Language (text-to-text, bilingual English-Chinese translation), code (code generation), scientific literature comprehension, summarizing, and report generation**
Amazon	AWS Artificial Intelligence (x 13 free products) **Polly, Transcribe, Lex, CodeWhisperer,**	**Language (text-to-speech, speech-to-text, conversational text + speech,** translation), automated data extraction and analysis, healthcare (data science), **code (code generation)**/devOps, search, computer vision, personalization
	Amazon Bedrock*	Centralized API for industry-leading GAI APIs across different companies
Alibaba	[Zhipu investor]	
Meta	**Llama 2**	**Text-based chatbot**
	Meta AI	Ongoing research brain activity-to-image,
Tencent	[Zhipu investor]	
Google	**Bard**	**Text-based chatbot**
Baidu	**Ernie AI**	**Text-based chatbot, hosted on open-source development platform Paddle Paddle**
	Wenku	**Content retrieval and creation, summary + assistant chatbot**
Microsoft	**Bing Chat**	**Text-to-text chatbot as Bing web browser plugin**
	OpenAI investor	**Text-based chatbot**
	Xiaoice	**Text-based chatbot**
KingSoft	**WPS AI**	**Text-based chatbot integrated in all its software apps**
Intel	**Developer Cloud**	**GPU and cloud-based platform to run AI applications**
SMIC**	**Semiconductor chip**	**Run AI applications, In partnership with Huawei**
TSMC*	**SC Chip**	**Run AI applications**
	Intelligent Fab	**System-of-systems-style Fabrication model integrating AI/ML and traditional tech**
IBM	**(see below)**	**(see below)**
IBM China	**DeepQA's watsonx**	**AI data platform, app distribution, text-based chatbot assistant; additional collaborations with Meta**

Company Considerations

*Components of the US military currently hold a subscription for SageAI, which centralizes different companies' chatbot APIs in a similar way to Amazon Bedrock. The difference is that, while Bedrock is for developers using APIs, SageAI's AskSage is a chatbot (not for development).

** 10 other semiconductor manufacturers of significant size operate out of China, yet Semiconductor Manufacturing International Corporation is the largest on in the country and the fifth largest globally.

***Taiwan Semiconductor Manufacturing is the largest global semiconductor manufacturer and services globally, including both the US and China

Companies bordered in red service both the US and China.

Companies bordered in brown are strongly based in education.

Companies highlighted in pink may be of particular interest to AETC

Bonus companies to watch: Mistral AI (France)

APPENDIX B: Intellectual Property

China leads the US in patents, sometimes manyfold, in the space of GAI. Numbers calculated from Google Patents search up through Oct 2023[6]

It is also worth noting that patents rarely reveal the most updated research in real time, but rather some of the most recent *protected* IP. Sometimes companies decide to not publish at all or publish with greater discretion. Nevertheless, the below table outlines three top US GAI patent filers and their roughly analogous Chinese companies, respectively.

Generative AI patents published

company	count	% granted during/after 2020
Amazon	68	96
Meta, Facebook	51/57, 24/27 (none in 2020)	89
Google	69	81
Alibaba	101	94
Tencent	460	95
Baidu	345	96

Whereas the Chinese-US gap in publication was relatively small before 2020, and although both countries raced to publish after 2020, China's companies overall accelerated at 3-4x the patent efficiency of the US. Research suggests Chinese universities also dominate patent fields over US universities.

APPENDIX C: Chinese development design considerations:

1. Baidu PaddlePaddle:
    a. Ernie Bot: different from ChatGPT:
        i. Trained on unstructured text
        ii. Knowledge graph
        iii. Image generation
    b. Wenxin Yige: complex art comprehension and completion
    c. Wenxin baizhong: simplify creation of GAI search engines by industry
2. Classroom:
    a. Surveillance from non-generative AI tech: attention span headband
    b. Squirrel AI: tutor bot

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2 https://sghexport.shobserver.com/html/baijiahao/2023/12/05/1196748.html . The latter has been particularly useful for English and other foreign language instruction.
3 https://teaching.cornell.edu/sites/default/files/2023-08/Cornell-GenerativeAIForEducation-Report_2.pdf
4 The nation's top-ranked province on the comprehensive development index of basic education informatization, http://jyt.zj.gov.cn/art/2023/7/19/art_1532773_58937425.html
5 http://jyt.zj.gov.cn/art/2023/3/20/art_1532836_58939666.html
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15 http://www.81.cn/yw_208727/16270005.html
16 https://www.nesdnow.navy.mil/docs/NESD%20Amelia%20Fact%20Sheet_v1.1_Dec2022_Final.pdf
17 Although we were unable to identify a Chinese military LLM or chatbot platform, an open-source

chatbot platform (by Aimoshu) accessing both US and Chinese LLMs appeared during some military-related Baidu searches. Website: chat.aimoshu.cc

18 http://www.81.cn/yw_208727/16270227.html

19 https://myjob.dlmu.edu.cn/search/list/do123/myjob.dlmu.edu.cn/domain/myjob/keyword/ai/page/2

20 https://ai.usajobs.gov/search/results/?cmco=AI&s=relevance&sd=asc&p=1&k=artificial%20intelligence